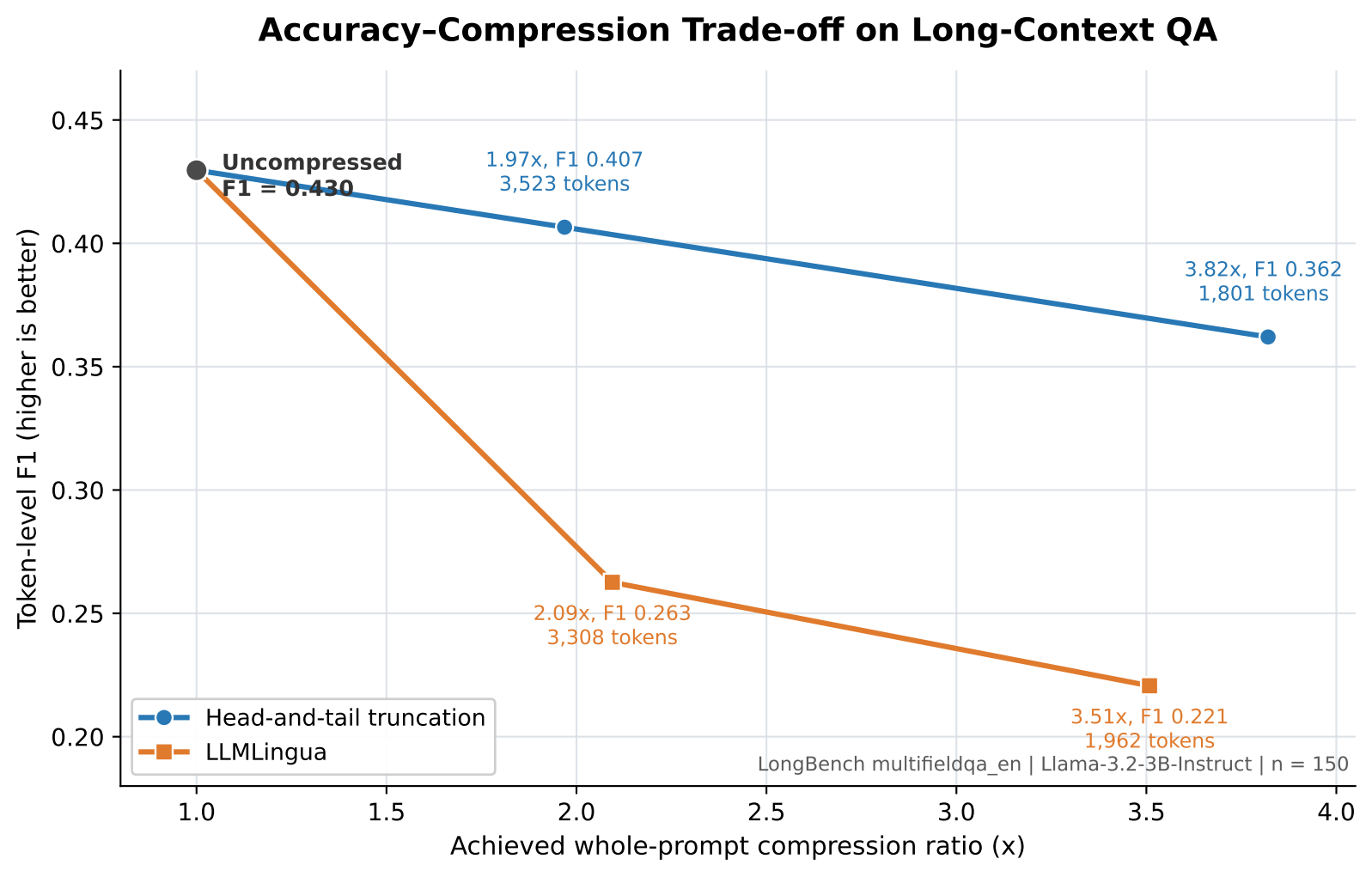
Accuracy–Compression Trade-off on Long-Context QA
Token-level F1 (higher is better)
Achieved whole-prompt compression ratio (x)
0.45
0.40
0.35
0.30
0.25
0.20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Uncompressed
F1 = 0.430
1.97x, F1 0.407
3,523 tokens
3.82x, F1 0.362
1,801 tokens
2.09x, F1 0.263
3,308 tokens
3.51x, F1 0.221
1,962 tokens
Head-and-tail truncation
LLMLingua
LongBench multifieldqa_en | Llama-3.2-3B-Instruct | n = 150